

# Project Title: Kolkata Scooty Bike Training – Slot Booking Web Application

## Objective

To create a responsive web application for customers to book 30-minute scooty training slots between 9:00 AM and 11:00 AM each day. The admin can manage trainers, customers, and slot availability dynamically.

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## Technology Stack

| Layer                    | Technology Used                                                      | Purpose                                                                    |
|--------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Frontend                 | Angular 17                                                           | UI development, responsive design, and dynamic routing                     |
| Backend                  | FastAPI (Python) ( <i>Preferred for performance and simplicity</i> ) | API creation, authentication, booking logic                                |
| Database                 | PostgreSQL                                                           | Data storage for users, trainers, slots, and booking history               |
| Authentication           | Google OAuth 2.0                                                     | Easy sign-in for customers                                                 |
| Hosting / Deployment     | Google Cloud Platform (GCP)                                          | Used for deploying both frontend (Angular) and backend (FastAPI) services. |
| Version Control          | GitHub                                                               | Collaboration and code maintenance                                         |
| Optional Add-on (Future) | Humanoid AI Assistant (Chat-based help for booking &                 | Interactive chatbot integration ( <i>Optional</i> )                        |

## System Overview

### Customer Web App (Angular UI)

#### Main Features:

- Customers can sign in via Google or OTP.
- Slot availability displayed in card/thumbnaill view (mobile-friendly).
- Only slots between 9 AM to 11 AM are visible.
- Each slot is 30 minutes.
- If a slot is booked, the next available slot for that same trainer appears automatically.
- Booking confirmation with email/SMS notification.

#### Customer Modules:

- Dashboard (Welcome + Next Booking)
  - Book Slot (Available trainers & slots)
  - Booking History
  - Profile (Update contact number)
  - Logout
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### Admin Panel

Technology: Angular (separate admin route) + FastAPI admin APIs

## Modules:

- Trainer Management: Add/Edit/Delete trainers and set their available slots.
- Slot Control: Open/Close any slot for specific trainers manually.
- Customer Management: View registered users and booking history.
- Booking Analytics Dashboard:
  - Total bookings/day
  - Slot utilization percentage
  - Trainer performance
- Notifications: Send custom notifications (email/SMS) to customers.

## UI Design Notes:

- Dashboard cards with key stats
- Filters (by trainer/date)
- CRUD tables with pagination
- Admin login with JWT authentication



## Timeline (28 Days / 14 Hrs per day)

| Phase   | Task                                                   | Duration |
|---------|--------------------------------------------------------|----------|
| Phase 1 | Requirement Analysis, UI Wireframes, DB Schema Design  | 3 Days   |
| Phase 2 | Angular Frontend (Customer Side) – Booking Flow & Auth | 8 Days   |

|         |                                                             |        |
|---------|-------------------------------------------------------------|--------|
| Phase 3 | FastAPI Backend – API Endpoints, Google Auth, Booking Logic | 7 Days |
| Phase 4 | Admin Panel Development (Trainer, Slot, Reports Modules)    | 6 Days |
| Phase 5 | Integration Testing (Frontend + Backend + Database)         | 2 Days |
| Phase 6 | UI Optimization, Deployment, Documentation                  | 2 Days |

Total Duration: 28 Days

Effort: ~14 hours/day

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## Database Design (PostgreSQL)

Tables:

- **users** → (id, name, email, phone, google\_id, created\_at)
- **trainers** → (id, name, specialization, status)
- **slots** → (id, trainer\_id, date, start\_time, end\_time, is\_booked)
- **bookings** → (id, user\_id, slot\_id, booking\_status, created\_at)

Constraints & Logic:

- Unique slot per trainer per time range.
  - Cascade booking updates.
  - Slot auto-refresh when one closes.
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## Future Enhancements

- Add Humanoid Assistant (optional AI module)
    - chat-based slot booking help.
    - FAQs and directions.
  - Add Payment Gateway (Razorpay/Stripe).
  - Trainer rating and feedback system.
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## Summary

The project delivers a complete end-to-end booking platform for scooty training with strong admin control, responsive UI, and secure authentication. Built for scalability, the system is ready for integration with advanced modules like AI assistants and payment gateways.

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## Deliverables

- Angular Frontend (Customer + Admin)
  - FastAPI Backend APIs
  - PostgreSQL Database Schema
  - Google Auth + GMAIL OTP Integration
  - Deployment Setup Documentation
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## Optional Humanoid Assistant (Future Ready)

Although not included now, the architecture supports integrating a humanoid or chatbot assistant for booking guidance, trainer communication, and FAQs.